

User run command (application begins)

- Launch ollama-based llm api

URL search & save

- Read data/company_list.csv
- Search for linkedin jobposts using
- Validate each url and save it into data/company_list.csv with a new column career websites. (optional)

LLM agent

- Access to given career websites and each job posts in the website
- MCP tools
 - FETCH_HTML: Read each job post in html and convert into readable texts (and usable to llm prompt) and return the results.
 - FETCH_PDF (optional): Read data/full_resume.pdf and convert the content into proper structured texts.
 - SAVE_RESULT: collect all features and save them into data/job_list.csv
- LLM api (initial screening): Read each description (saved in outputs/JobPosts_{position abbr}.csv and analyze the following for initial check. Look for easy key feature checks. Save filtered jobposts into outputs/FilteredJobPosts_{position abbr}.csv.
 - Job title: should not include 'staff', 'principal', 'lead', 'manager', 'infrastructure', 'cuda', 'Researcher', but 'technical staff' is allowed.
 - Years of experience: should not be more than 7.
- LLM api (llm-based screening): read data/full_resume.pdf and each description from initially-screened outputs/FilteredJobPosts_{position abbr}, put them into llm prompt for ollama api and calculate similarity
 - (Optional) Embedding api: use document an embedding model to calculate embeddings from my resume and job descriptions and produce similarity
 - Based on similarity scores, write those jobs above a pre-defined threshold score into outputs/TargetJobPosts_{position abbr}.csv with columns: jobId, company_name, similarity score, job descriptionURL, job descriptions.
 - JobID: {company_name}_{position abbr}_{unique 4 digits}

LLM agent

- Given job descriptions from and original tex file, tailor resume to each job description and save it to pre-defined dir with naming rules.
- MCP tools
 - HANDLE_TEX: read my resume tex file and save a tailored tex file to tailored_resumes/{position_abbr}/JobID/...
 - CONVERT_TEX_TO_PDF: read generated tex files, convert them into pdf, write them into the same dir.
- LLM api: Read each description from outputs/TargetJobPosts_{position abbr}.csv and read data/full_resume.tex. Then, embed them into prompt for creating a tailored resume in latex format. Save the generated answer into tex and pdf.